

Scalability & Future Plan

Date	1 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the **Smart Lender – Loan Approval Prediction System** can be scaled to support a larger number of users, higher volumes of loan applications, and additional intelligent features. The project currently predicts loan approval using a trained Machine Learning model integrated with a Flask web application. Future enhancements will improve performance, security, scalability, and usability while making the system suitable for deployment in real banking environments.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only a limited number of concurrent users	Performance decreases during heavy traffic	High
2	Uses a local pickle model without cloud deployment	Limited accessibility and scalability	High
3	No user authentication or loan application history	Reduced security and user management	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask server	Deploy using cloud platforms with load balancing
2	Data Storage	Local files	Store customer data in MySQL/PostgreSQL or MongoDB
3	Performance	Single ML prediction process	Optimize model and use caching/API services
4	Security	Basic web application	Add user authentication, HTTPS, encryption, and role-based access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase1	User Login & Registration	1 Month	Secure user access
Phase 2	Cloud Deployment (AWS/Azure/Render)	2 Months	High availability and scalability
Phase 3	Loan Status Tracking & Dashboard	3 Months	Better customer experience
Phase 4	AI-based Loan Recommendation & Document Verification	4–6 Months	Improved automation and prediction accuracy